

CS Capstone Design

Product Demonstration Grading Sheet (100 pts)

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PROJECT: REACH

DEVELOPMENT CYCLE: ☐ Alpha Demo I ☐ Alpha Demo II ☒ Acceptance Demo

DATE: 4/28/2026

MENTOR: Bailey Hall

CLIENT: Dr. Lerner and Dr. da Cunha

1. Overview

The purpose of the Product Demonstration Flight Plan is to define what functionality is planned, what functionality will be demonstrated in the current development cycle, and how success will be evaluated.

This document serves three roles:

1. A **planning artifact**, showing how the full product scope is distributed across development cycles
2. A **demonstration guide**, outlining how the current cycle's functionality will be presented through realistic user flows
3. An **evaluation aid**, defining acceptance criteria used by the mentor and client to assess completeness and correctness

The Flight Plan is prepared by the team, reviewed and approved by the mentor, and used during the live demonstration for evaluation and feedback.

2. Product Scope and Development Cycles Overview

This section presents the full set of user stories or use cases for the product and how they are distributed across the three development cycles. The intent is to make clear:

- What functionality exists in the overall product, corresponding to your full prioritized list of functional requirements
- What is included in each development cycle
- How incremental delivery builds toward the final system

2.1. Backlog Overview by Development Cycle

Alpha Demo I

ID	User Story / Use Case	Acceptance Criteria (Summary)
US-1	As a user, I want to launch a simulated environment so that I can observe a system operating in a virtual setting.	MuJoCo scene loads without errors; the robotic arm appears correctly positioned relative to the user model; simulation runs and renders consistently.
US-2		The arm responds to control inputs; joints move within defined limits; no instability or simulation crashes occur.

	As a user, I want the robot to execute basic joint movements so that I can verify control and kinematics.	
US-3	As a researcher, I want to train a reinforcement learning agent on a simple reaching task so that I can confirm the learning pipeline works end-to-end.	Training script runs successfully; reward values update during training; trained policy produces non-random arm motion toward a target.
US-4	As a developer, I want to visualize training and evaluation rollouts so that I can validate agent behavior qualitatively.	Evaluation script renders rollouts; end-effector trajectory is visible; logs confirm successful episode execution.

Alpha Demo II

ID	User Story / Use Case	Acceptance Criteria (Summary)
US-1	As a user, I want the robotic arm to reliably reach a defined target with full body motion enabled in all the joints.	The end-effector reaches within a defined distance threshold of the target in a majority of evaluation episodes.
US-2	As a researcher I want the framework to be easily adaptable to multiple robot types and scenes.	The framework has relatively easily swappable robots, and a new scene or robot can be uploaded and tested by a researcher with minimal technical skills needed.
US-3	As a researcher, I want the AINEX robot to be able to recognize a user gesture and report back to the user.	The ainex robot is able to use computer vision to determine if a human hand is in front of its camera and correctly identify a "high five" gesture.

Acceptance Demo - Current Cycle

ID	User Story / Use Case	Acceptance Criteria (Summary)
US-1	As an end user, I want the ainex robot to respond to a given gesture with a specific action group.	When the robot is presented with a high five, it should respond with a specified action.

US-2	As a researcher, I want the framework to optimally support various robot types and easily start the training of a robotic arm for a reaching task, and the rendering of the arm with and without the policy.	The file structure and YAML parameters are intuitive to configure and new robots can be easy to upload and get trained, and rendered, with or without their policy.
US-3	As a project stakeholder, I want documentation of system design and usage to be clear and well defined.	Documentation is well written and conducts users to install, update, and use the framework as intended.

3. Demonstration Walkthrough

For each user story or use case included in the current cycle, provide a demonstration walkthrough guided by acceptance criteria. The demo should follow a realistic user flow and show the functionality working end-to-end.

Use Case / User Story: US-1 Fist Bump Demo

User Role: End User

Goal: Demonstrate Gesture Recognition and Correct Reactive Behavior from the AiNex

Demonstration flow:

1. User Positions in front of the AiNex Robot
2. User gestures with a highfive to the AiNex
3. The Ainex detects the gesture internally or textually, giving feedback via the terminal window
4. The Ainex Robot responds with the appropriate action

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. The AiNex correctly detects the highfive gesture.
2. The AiNex correctly responds with the correct gesture response.

Evaluation and Comments (for mentors only):

- Convincingly demonstrated the functionality
- Other evaluative comments/notes

Use Case / User Story: US-2 Framework adaptability

User Role: Researcher

Goal: The framework is more easily able to support uploads, training, and rendering of different robot arms.

Demonstration flow:

1. A newly downloaded robot from the internet is given
2. Files are shown as uploaded to the project in the specified location
3. Using YAML config, the robot path is specified along with its other new parameters
4. The render scene script is ran to show the new scene renders with a universal script
5. Using YAML config, the arm can be trained without error
6. The render script for replaying a trained policy is given
7. The file structure and ease of use of the whole framework is then explained

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. Simulation opens for a specified scene from the YAML config
2. A training session is ran from the universal file
3. The rendering of the arm is successful
4. The file structure is intuitive and easier than the alternative of starting from scratch

Evaluation and Comments (for mentors only):

- Convincingly demonstrated the functionality
- Other evaluative comments/notes

Use Case / User Story: US-3 Impeccable Documentation

User Role: Researcher

Goal: The entire codebase is Clean and Intuitive

Demonstration flow:

1. The full frame work architecture is displayed and explained
2. Documentation and code quality is displayed
3. Neccessary documents and config files are also shown and explained
4. It is explained why the project is ready for delivery to the clients

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. The full codebase is cleanly and intuitively documented and structured
2. The final product is prepared for delivery to the clients

Evaluation and Comments (for mentors only):

- Convincingly demonstrated the functionality
- Other evaluative comments/notes

4. Out-of-Scope or Deferred Functionality

List any user stories or use cases planned in the approved flight plan for this cycle, but not demonstrated, along with their planned future cycle. This helps set clear expectations and avoids confusion during the demo.

Acceptance Demo – Current Cycle

ID	Reason Not Demonstrated	Planned Cycle

5. Known Limitations and Planned Follow-Up

Document any known limitations (if any), partial implementations, or issues observed during development or demonstration. Where possible, indicate planned mitigation or timing for resolution.

- Limitation / Issue:
- Impact:
- Planned Resolution (if known):

6. Approval

This Flight Plan has been reviewed and approved for use in the Alpha Demonstration.

Mentor Approval: Bailey Hall

Date: 4/28/2026